工作成果证明

1. 浅底波导场的声波传播 项目研究

A． 相关的一些论文发表

1. R. Gilbert &Zhongyan Lin, Fundamental singularity in a shallow ocean, Applicable Analysis,
68(1998), No. 1-2p87-107
2. R. Gilbert &Zhongyan Lin, Acoustic field in a shallow stratified ocean with a poro-elastic seabed, ZAMM. Z. angew. Math. Mech. 77(1997) 9, p677-688
3. R. Gilbert &ZhongyanLin, Underwater Acoustics, Generalized Analytic Functions, Kluwer Academic Publishers, The Netherlands, p215-228
4. R. Gilbert &Zhongyan Lin, Direct and inverse problems in ocean acoustics, Proceeding of the Second World Congress of Nonlinear Analysts, Part 3(Athens, 1996), Nonlinear Analysis, 30(1997), No. 3, p1535-1546
5. R. Gilbert &Zhongyan Lin, An acoustic inverse problem: numerical experiment, Journal of Computational Acoustics, Vol.3, No.3(1995) p229-240 (joint with Gilbert)
6. R. Gilbert &Zhongyan Lin, Acoustic waves in shallow inhomogeneous ocean with a layer of sediment, Acustica, Vol.82(1996), p729-737
7. R. Gilbert &Zhongyan Lin and James Buchanan, Acoustic waves in shallow inhomogeneous ocean with an interactive seabed, Proceedings of the Conference on Mathematical Modeling of Flow Through Porous Media,France,May1995
8. Zhongyan Lin & R. Gilbert, Numerical algorithm based on transmutation for solving inverse wave equation, Mathematical and Computer Modelling, Volume 39, Issue 13, June 2004, Pages 1467-1476
9. R. Gilbert &Zhongyan Lin, Scattering in a shallow ocean with an elastic seabed,

Journal of

Computational Acoustics , Journal of Computational Acoustics, Vol.5, No.4 (1997) 403

B． 项目组长 Robert R. Gilbert 和另一研究员 James Buchanan 教授的工作推荐信有很多提及工作成果

University of Delaware

January 24, 1999

ROBERT P. GILBERT
UNIDEL CHAIR FOR APPLIED ANALYSIS
DEPARTMENT OF MATHEMATICAL SCIENCES
UNIVERITY OF DELAWARE
NEWARK, DELAWARE 19716
e-mail gilbert@math.udel.edu
FAX 302-368-1835

(302) 831-2315

Department of Justice
Immigration and Nationalization Services

To Whom it May Concern:

Re: Letter of recommendation to grant Immigration to Dr. Zhongyan Lin under the Category of Professionals with Extraordinary Ability

This letter is written to support the application of Zhongyan Lin for immigration to the United States under a special category of professionals with extraordinary ability or outstanding researcher. I hold the Unidel Chair for Mathematics and Computer Science at the University of Delaware. I am the author of over 200 research papers, and the author of 6 research monographs. I am Editor-in-Chief of the two mathematics journals APPLICABLE ANALYSIS and COMPLEX ANALYSIS; moreover, I am on the editorial boards of 12 other research journals. I first met Lin when I was visiting China. He impressed me at that time with his sharp mind and we started a joint collaboration on a paper during my visit. I was so impressed with him that I recruited him right away to come and finish his Ph. D. with me at Delaware. He had participated in several important mathematical research and engineering design projects in China and had written several very good papers in that field before coming to work with me. We wrote one paper together on inverse problems before I started him on computational acoustics. Dr. Lin wrote an brilliant Ph. D. thesis under my direction on computational acoustics and inverse problems for mathematical physics. Dr. Lin and I collaborated subsequently on numerous research efforts and published jointly eight papers. Dr. Lin's research was mainly concerned with computational acoustics for shallow ocean with an interactive seabed, and the undetermined coefficient problems for partial differential equations. These are ill-posed problems, and very difficult. He is working in areas that are very important to industry and national security. The direct application of his work is for the detection of mines that might be placed in a harbor. The same technology developed in his research is of use in determining the nature of the sea-bed, which can be used in the many areas such as prospecting for off-shore petroleum and gas, and constructing drilling platforms. His research, which is related to non-destructive evaluation (NDE), has numerous commercial, defense, and health applications. For

example, non-destructive evaluation is being used to evaluate aged aircraft in the Air Force, and inverse scattering is of potential use in medical diagnoses. Lin has accomplished an immense amount in the area of computational acoustics. He is a great problem solver, in particular, in the area of partial differential equations involving inhomogeneous media and non-constant coefficients. This is because he has a solid and extensive training in mathematics, physics and computer science. He specializes in both theoretic approaches and numerical methods. He can design and implement the most efficient algorithms for many challenging problems. Because of his excellent work, now we have several effective ways for computing the acoustic field in a shallow ocean. This includes various types of sea-beds, with various sound sources, and also with objects embedded in the sea-bed. His research in this field has components which vary from abstract mathematics such as functional analysis to the implementation of these ideas in the terms of very large computer codes. His codes were run on the Cray at Pittsburgh and were very successful for determining unknown objects such as submarines and/or mines. Lin's research is on the cutting edge of modern computational acoustics, and Navy applications of these methods. His code for a full three-dimensional acoustic field generated by scattering an illuminating signal is state of the art. There are few scientists in the world today who have the command not only of an exquisite knowledge of scientific computation but also the command of abstract mathematics to permit a break through in the field of inverse acoustic problems in an ocean environment. Dr. Lin's work in inverse problem is also outstanding. His recent paper "On the determination of radially dependent Lame coefficients" has caught great attention among the researches in this field. This long paper published in SIAM Journal on Appl. Math., one of the best journals in the field, proves a very important theorem in the area of non-destructive evaluation and the method he established has a potential for solving numerous problems in the field. His results is one of best results on inverse problem in recent years. Because Dr. Lin is a very active researcher, he was asked to review submitted papers for publication and international conference many times. Lin also has an advanced degree in Computer Science. His combined knowledge of acoustics, and computer science and information systems make him a great asset to our country for the development of technology and hence also the economy of our nation as a whole. Dr. Lin is an excellent applied mathematician and outstanding researcher. And he is clearly the best Ph.D. student I have had in 40 years of graduate teaching! At this time Dr. Lin is employed in the aeronautic and automatic control industry and his knowledge of inverse problems and acoustics can be easily transferred to identification of objects, collision avoidance, noise control and other automatic control, which of course are of great importance to that industry. It is indeed rare when a researcher such as Zhongyan Lin appears on the scene. Not only is he a highly gifted, and well trained computational-mathematician, he has, an extraordinary knowledge of mathematical physics. Currently, I am particularly interested in collaborating with him to solve more inverse scattering problems. This efficient solving of this problem is of great importance for National security. There will always be a contest between potential aggressors and the scientist who strive to protect our shores. It is necessary that we keep a monopoly of those scientists here in our country. With this object in mind, I recommend that you give serious attention to considering Dr. Zhongyan Lin for special waiver of Labor Certification. Dr. Lin is an outstanding scientist and is

Professor James Buchanan:

DEPARTMENT OF THE NAVY
UNITED STATES NAVAL ACADEMY
ANNAPOLIS, MARYLAND 21402-5000

572 Holloway Road
Annapolis MD 21402

18 February 1999

Department of Justice
Immigration and Naturalization Services

Subject: Recommendation to grant immigration to Dr. Zhongyan Lin under Professionals with Extraordinary Ability or Outstanding Researchers

To whom it may concern,

I am writing this letter in support of the application of Zhongyan Lin for immigration to the United States under a special category of National Interest Waiver for professionals holding advanced degrees. I am a professor of mathematics at the United States Naval Academy. I am the author of a number of papers in underwater acoustics and the mathematical theory of elasticity as well as a research monograph in partial differential equations and a textbook in numerical analysis.

I became acquainted with Dr. Lin when he was a graduate student of my long term collaborator Dr. R.P. Gilbert of the University of Delaware. I have had occasion to discuss his work on the effect of the composition of the seabed on acoustic transmission in shallow water, a subject of mutual interest. I invited him to the Naval Academy to speak on his work on inverse problems last year. Additionally I have refereed one of his papers for publication.

Dr. Lin's work is first rate. He made significant breakthroughs in the mathematical theory of acoustic transmission over an elastic seabed, in particular discovering an inner product that will permit more rapid calculation of the acoustic field arising from sound scattering off an object in a shallow ocean. This discovery should may also facilitate the extension of current methods for the inverse problem of reconstructing the shape of an object from the sound waves scattered off of it. Solution of such problems in the presence of a seabed are currently of great interest to the Navy since it is anticipated that in the near future many naval conflicts will be conducted in littoral areas.

I am also impressed by his work on other inverse problems such as the determinations of the coefficients of the partial differential equation governing an elastic body from measurements made on it surface. Inverse problems such as Dr. Lin works on are of great importance in numerous fields such as bio-medicine and seismology, as well of course, as underwater acoustics and the underlying mathematical and computational techniques tend to be the same irrespective of the field of application.

I should also mention that Dr. Lin holds a master's degree in computer science as well as a doctorate in mathematics. Almost all problems of importance in applied mathematics have a significant computational component. In my experience the most effective solutions are

obtained by people who have command of both the analytical techniques and the computational algorithms involved in the problem. Unfortunately people with strong backgrounds in both analysis and computation are uncommon. Dr. Lin's proficiency in both makes him a valuable asset.

In summary Dr. Lin possesses valuable and rare skills which are of importance to the United States both militarily and commercially. I strongly support his application for immigrant status under the National Interest Waiver.

Sincerely,

James L. Buchanan
Professor, Mathematics Department
Tel: (410) 293-6726
Fax: (410) 293-4883

2. 径向对称相关球面未知弹性系数的反问题

成果发表的 SIAM Appl. Math. 这是应用数学里最著名的刊物，我的 30 页的论文是最长的：摘要见 https://epubs.siam.org/doi/abs/10.1137/S0036139996303038

全文见 https://zhongyanlin.github.io/site/publication/siam.pdf

首页的底部提及研究基金来源

SIAM J. APPL. MATH.
Vol. 58, No. 3, pp. 875–903, June 1998

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008

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ON THE DETERMINATION OF RADIALLY DEPENDENT LAMÉ COEFFICIENTS*

ZHONGYAN LIN†

Abstract. In this paper, we prove that one can determine radially dependent Lamé coefficients λ and μ of an isotropic elastic ball by choosing only two correspondences in the Dirichlet-to-Neumann map, i.e., twice making measurements of displacement and stress on the surface of the ball. This is a one-dimensional, multiunknown, undetermined coefficient problem for a system of partial differential equations. We use a matrix form of the transmutation mapping for solving this problem. The procedure for proving the uniqueness based on transmutation can be viewed as a computational algorithm. Also the method we develop has the potential of solving other one-dimensional, multiunknown inverse problems.

Key words. inverse problem, elasticity, elliptic system, one dimension, Lamé coefficients, computer algebra, matrix, transmutation, moments, hyperbolic system

AMS subject classifications. 35R30, 73C02

PII. S0036139996303038

1. Introduction to the inverse problem and the main result. The problems of determining the coefficients in partial differential equations are important and interesting inverse problems. The coefficients often describe the physical properties of the region to be detected. Most studies of undetermined coefficient problems involve a single unknown coefficient in a scalar equation such as $u_{tt} = u_{xx} + a(x)u(x)$ or $\mathrm{div}(\gamma(\vec{x}) \bigtriangledown u) = 0$. Recently, inverse problems for systems of equations are being considered. In fact, for many problems in elastic and electromagnetic inhomogeneous media, only by restricting the coefficients do we get a single potential equation. For example, if only one coefficient is variable while others are constant, we can decouple the system of equations; or, on the other hand, when we set some coefficients to be zero, we may get a single acoustic-type equation. Otherwise we have to deal with systems of equations and identify simultaneously multiple coefficients.

Since for radially symmetric objects the inverse problems for a single potential equation have been well studied [12, 17, 6, 4], in this paper we consider an inverse problem for a system of equations with radially dependent coefficients. The problem is to determine the radially dependent Lamé coefficients of an elastic ball by the measurements of stress and displacement taken on the surface. To introduce this problem more precisely, first we quote from [1] the following formulation for the problem of determining variable elastic coefficients by boundary measurements.

Let $\Omega \subset \mathbb{R}^n$ be a domain with smooth boundary $\partial\Omega$ and filled with a linear, inhomogeneous elastic medium. The deformation of Ω due to the given displacement vector $\vec{f} \in C^\infty(\partial\Omega)$ is expressed by the following Dirichlet boundary value problem in terms of displacement vector $\vec{u} = \vec{u}(x)$:

$$L_C\vec{u} = \mathrm{div}\sigma(\vec{u}) = 0, \quad \in \Omega, \tag{1.1}$$

$$\vec{u}\big|_{\partial\Omega} = \vec{f}, \tag{1.2}$$

*Received by the editors May 6, 1996; accepted for publication (in revised form) November 26, 1996; published electronically April 3, 1998. This research was supported by a fellowship from the Department of Mathematical Sciences, University of Delaware.
http://www.siam.org/journals/siap/58-3/30303.html

†Department of Mathematical Sciences, University of Delaware, Newark, DE 19716 (zlin@math.udel.edu).

3. 基于网络的教学过程管理和评估管理系统

系统的谷歌云安装见 https://edqaa.com

中文 https://edqaa.com/tealeaman/index.jsp?orgnum=65536

此次申请就是来探索请天宸启桦合作推向宁波（进而全球）高教市场。

以下是一个节目截图。特拉华州立大学高效教学助理校长的推荐信。

Dr. David Donald Pokrajac

Phone: 302-382-0381

Email: majstorpoki@gmail.com

Re: Recommendation Letter for Dr. **Zhongyan Lin**

Dear faculty searching committee:

It is my pleasure to recommend Dr. Zhongyan Lin for the faculty position that is available at your institution in the field of computer information science and data analytics.

I worked with Dr. Lin for 16 years at Delaware State University (DSU) before I joined my current company in 2018. I know Dr. Lin very well. He is a very friendly person, who works very hard in teaching, research and software development. For many years, he taught the most course sessions and most students in the department of computer and information sciences at DSU.

From his fully-verified publication list (https://zhongyanlin.github.io/site/publication), we can see that he had done decent research work in the field of partial differential equation and inverse problems, published papers in some of most prestigious journals such as SIAM J. Applied Math.

I worked with Dr. Lin on a project of data analytics titled "Correlation of Risk of Breast Cancer with density of breast density" funded by Army research office and Medical School of University of Pennsylvania for a period of three years . We together processed 1.5 Terabytes of clinical data, including image data, text information and numerical data. We performed analysis and completed reports for the submission to the funding agencies. Dr. Lin wrote programs, created and maintained an online database and produced all queries to generate reports.

Before this project, Dr. Lin also worked on another data analytics project funded by Army research lab in 2008. That project was about statistical modeling (the application of maximum likelihood method to determine critical penetration distance in armor).

Dr. Lin has been an early proponent of utilizing computer technology for class management, students' grading and computer-assisted education. As such, he preceded for years the current trends in on-line and hybrid learning. From his annotated list of software of public interests that he developed (https://zhongyanlin.github.io/site/), we can see that he recently published three apps in the google play store on teaching technologies and data analytics using artificial intelligence. They are:

- Grading Scanner: https://play.google.com/store/apps/details?id=com.systemsonweb.grading,
- Enquire-response: https://play.google.com/store/apps/details?id=com.systemsonweb.response,
- Facial attendance: https://play.google.com/store/apps/details?id=com.systemsonweb.attendance.

I strongly believe that Dr. Lin's background in both computer and information science and mathematics makes him an excellent candidate for teaching position in the field of computer information science or data analytics.

Sincerely,

David Donald Pokrajac, Ph.D.

Former Associate Vice President

of Institutional Effectiveness

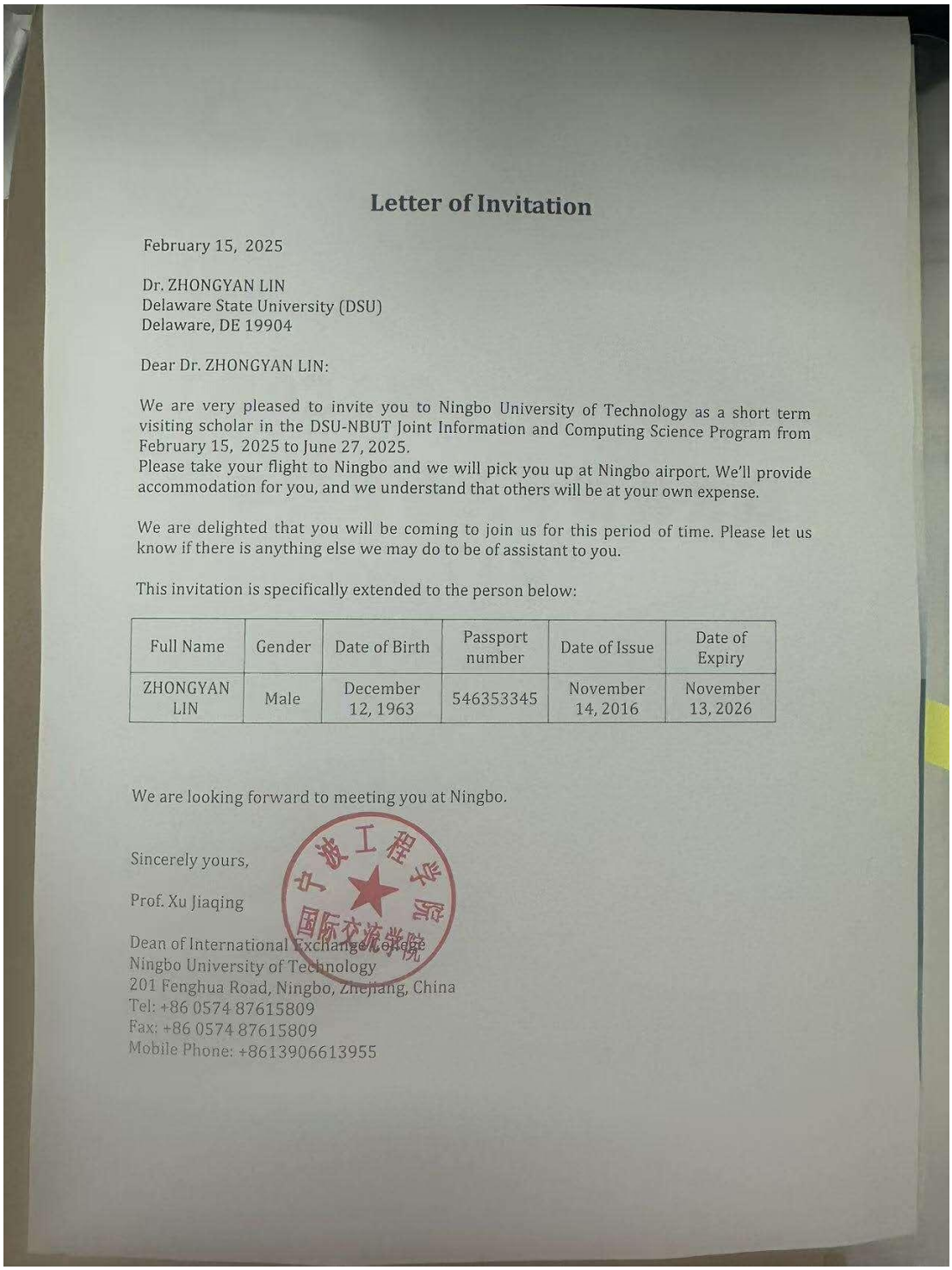
at Delaware State University

Visiting professor,

University of Nis, Serbia

4. 开发、教学、质量管理特拉华州立大学中国校区.

参与特拉华州立大学在中国开设海外合作项目（这些合作项目每年为雇主带来数百万美元的创收）。证据：受邀到宁波工程学院教学，排课和分数簿。



Letter of Invitation

February 15, 2025

Dr. ZHONGYAN LIN
Delaware State University (DSU)
Delaware, DE 19904

Dear Dr. ZHONGYAN LIN:

We are very pleased to invite you to Ningbo University of Technology as a short term visiting scholar in the DSU-NBUT Joint Information and Computing Science Program from February 15, 2025 to June 27, 2025.
Please take your flight to Ningbo and we will pick you up at Ningbo airport. We'll provide accommodation for you, and we understand that others will be at your own expense.

We are delighted that you will be coming to join us for this period of time. Please let us know if there is anything else we may do to be of assistant to you.

This invitation is specifically extended to the person below:

Full Name	Gender	Date of Birth	Passport number	Date of Issue	Date of Expiry
ZHONGYAN LIN	Male	December 12, 1963	546353345	November 14, 2016	November 13, 2026

We are looking forward to meeting you at Ningbo.

Sincerely yours,

Prof. Xu Jiaqing

Dean of International Exchange College
Ningbo University of Technology
201 Fenghua Road, Ningbo, Zhejiang, China
Tel: +86 0574 87615809
Fax: +86 0574 87615809
Mobile Phone: +8613906613955

宁波工程学院 国际交流学院

特拉华州立大学排课单显示我（下图右上角）下学期被排 6 个班的课

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RETURN TO MENU SITE MAP HELP EXIT

Faculty Detail Schedule

D10059906 Zhongyan Lin
Fall 2025
Aug 12, 2025 02:12 pm

ELEMENTS OF COMPUTER PROGRAMMING II - 15236 - CSCI 121 - N1

Status:	Active
Available for Registration:	Apr 01, 2025 - Sep 04, 2025
College:	Agriculture, Science & Tech.
Department:	Computer and Info Sciences
Part of Term:	1
Course Credits:	4.000
Course Levels:	Undergraduate
Campus:	Ningbo University
Override:	No
Syllabus:	Add
Rosters:	Classlist
Office Hours:	Add

Enrollment Counts

	Maximum	Actual	Remaining
Enrollment:	60	0	60
Cross List:	0	0	0

Scheduled Meeting Times

Type	Time	Days	Where	Date Range	Schedule Type	Instructors
Class	TBA		TBA	Aug 25, 2025 - Dec 12, 2025	Lecture	Zhongyan Lin (P), Marwan F Rasamny

ELEMENTS OF COMPUTER PROGRAMMING II - 15237 - CSCI 121 - N2

Status:	Active
Available for Registration:	Apr 01, 2025 - Sep 04, 2025
College:	Agriculture, Science & Tech.
Department:	Computer and Info Sciences
Part of Term:	1
Course Credits:	4.000
Course Levels:	Undergraduate
Campus:	Ningbo University
Override:	No
Syllabus:	Add
Rosters:	Classlist
Office Hours:	Add

Enrollment Counts

	Maximum	Actual	Remaining
Enrollment:	60	0	60
Cross List:	0	0	0

Scheduled Meeting Times

Type	Time	Days	Where	Date Range	Schedule Type	Instructors
Class	TBA		TBA	Aug 25, 2025 - Dec 12, 2025	Lecture	Zhongyan Lin (P), Marwan F Rasamny

PRINCIPLES OF PROGRAMMING LANGUAGES - 15623 - CSCI 380 - N1

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Status:	Active
Available for Registration:	Apr 01, 2025 - Sep 04, 2025
College:	Agriculture, Science & Tech.
Department:	Computer and Info Sciences
Part of Term:	1
Course Credits:	4.000
Course Levels:	Undergraduate
Campus:	Ningbo University
Override:	No
Syllabus:	Add
Rosters:	Classlist
Office Hours:	Add

Enrollment Counts

	Maximum	Actual	Remaining
Enrollment:	60	0	60
Cross List:	0	0	0

Scheduled Meeting Times

Type	Time	Days	Where	Date Range	Schedule Type	Instructors
Class	TBA		TBA	Aug 25, 2025 - Dec 12, 2025	Lecture	Zhongyan Lin (P), Marwan F Rasamny

PRINCIPLES OF PROGRAMMING LANGUAGES - 15623 - CSCI 380 - N1

Status:	Active
Available for Registration:	Apr 01, 2025 - Sep 04, 2025
College:	Agriculture, Science & Tech.
Department:	Computer and Info Sciences
Part of Term:	1
Course Credits:	3.000
Course Levels:	Undergraduate
Campus:	Ningbo University
Override:	No
Syllabus:	Add
Rosters:	Classlist
Office Hours:	Add

Enrollment Counts

	Maximum	Actual	Remaining
Enrollment:	60	0	60
Cross List:	0	0	0

Scheduled Meeting Times

Type	Time	Days	Where	Date Range	Schedule Type	Instructors
Class	TBA		TBA	Aug 25, 2025 - Dec 12, 2025	Lecture	Zhongyan Lin (P), Marwan F Rasamny

PRINCIPLES OF PROGRAMMING LANGUAGES - 15624 - CSCI 380 - N2

Status:	Active
Available for Registration:	Apr 01, 2025 - Sep 04, 2025
College:	Agriculture, Science & Tech.
Department:	Computer and Info Sciences
Part of Term:	1
Course Credits:	3.000
Course Levels:	Undergraduate
Campus:	Ningbo University
Override:	No

bnrhvprod-ssb.desu.edu/PROD/bwlkifac.P_FacSched

Syllabus:	Add
Rosters:	Classlist
Office Hours:	Add

Enrollment Counts

	Maximum	Actual	Remaining
Enrollment:	60	0	60
Cross List:	0	0	0

Scheduled Meeting Times

Type	Time	Days	Where	Date Range	Schedule Type	Instructors
Class	TBA		TBA	Aug 25, 2025 - Dec 12, 2025	Lecture	Zhongyan Lin (P), Marwan F Rasamny

ANALYSIS OF ALGORITHMS - 15629 - CSCI 485 - N1

Status:	Active
Available for Registration:	Apr 01, 2025 - Sep 04, 2025
College:	Agriculture, Science & Tech.
Department:	Information Technology
Part of Term:	1
Course Credits:	3.000
Course Levels:	Undergraduate
Campus:	Ningbo University
Override:	No
Syllabus:	Add
Rosters:	Classlist
Office Hours:	Add

Enrollment Counts

	Maximum	Actual	Remaining
Enrollment:	50	0	50
Cross List:	0	0	0

Scheduled Meeting Times

Type	Time	Days	Where	Date Range	Schedule Type	Instructors
Class	TBA		TBA	Aug 25, 2025 - Dec 12, 2025	Lecture	Zhongyan Lin (P), Marwan F Rasamny

ANALYSIS OF ALGORITHMS - 15630 - CSCI 485 - N2

Status:	Active
Available for Registration:	Apr 01, 2025 - Sep 04, 2025
College:	Agriculture, Science & Tech.
Department:	Information Technology
Part of Term:	1
Course Credits:	3.000
Course Levels:	Undergraduate
Campus:	Ningbo University
Override:	No
Syllabus:	Add
Rosters:	Classlist
Office Hours:	Add

Enrollment Counts

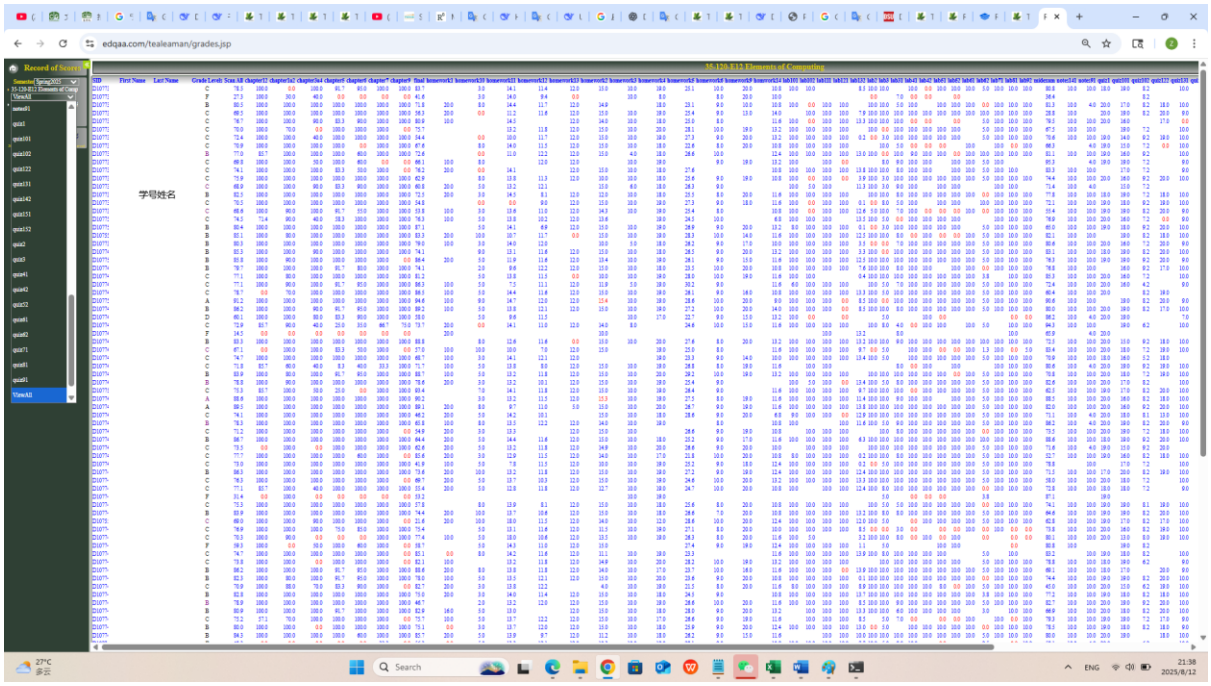
	Maximum	Actual	Remaining
Enrollment:	50	0	50
Cross List:	0	0	0

的比赛

Search

系统使我每学期能教 6 个班，每班近 50 人，每学生提交平 55 项课业。 产生百万美元的收益。这些数据用于产生质量评估报表，使项目的教育部审查年年通过

这是一个班的分数簿部分



5. 开源软件开发

和其他职业软件工程师，我也有 github 托管大开源软件. 3 兆源代码

https://zhongyanlin.github.io/site